Week 7 Financial News and Handling Text-Based Data

FINS5545: Financial Market Data Literacy

Introduction to Unstructured Text Data

Core Goals for This Course

By the end of the next four weeks, you will be able to...

1. **Work with text in `pandas`.** Load unstructured text into a dataframe and handle it beside your price and return data.

2. **Pre-process text with standard tools.** Use Python natural language processing (NLP) packages — `nltk`, `spaCy`, `scikit-learn` — to tokenise, remove stop words, and clean text quickly for analytics.

3. **Turn text into sentiment scores.** Apply rules- and lexicon-based methods, **VADER** and **TextBlob**, to transform headlines into a numeric tone score.

4. **Extend a lexicon with AI.** Use an AI agent to add finance-specific words to the VADER and TextBlob lexicons, so the score fits market language.

What Is Unstructured Text Data?

Structured data has a fixed structure — rows and columns, like the prices and returns from Weeks 1 to 5. **Unstructured data** does not. Text is the main example — it has no fixed schema, and every document is a different length.

Source	Example
news articles	"Is AbbVie's Stock a Buy?"
social media	posts on X, Reddit, StockTwits
company filings	a 10-K annual report
central bank	a Federal Reserve (FOMC) policy statement
earnings calls	a transcript of management's remarks

Working with this text is called **natural language processing (NLP)**. The goal is to put structure on the unstructured, turning words into numbers we can analyse.

Luckily for us, we can store text in a structured dataframe.

Text-Data Glossary (1) — Units and Counts

Keep this handy — these are the words commonly used in NLP.

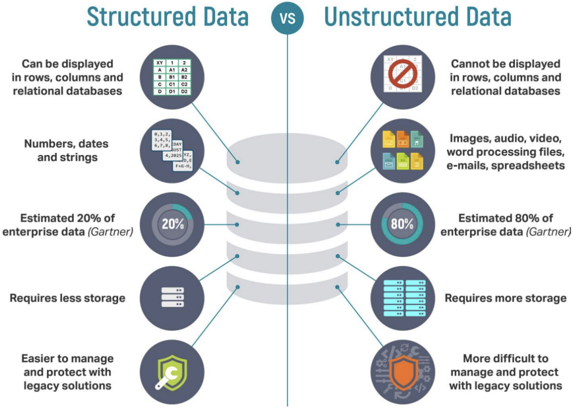
Term	Plain meaning
corpus	the whole collection of text you study — here, all your headlines
document	one unit of text in the corpus — here, one headline (one row)
token	one unit of text after splitting — in this course, one word
tokenise	split text into tokens, lower-cased runs of letters (*Is AbbVie's Stock a Buy?* → `is abbvie's stock buy`)
word count	the number of word tokens in a document (`n_words`)
vocabulary	the set of distinct words across the corpus
stop words	very common words (`the`, `is`, `of`) dropped only when counting frequent words
top terms	the words that appear most often for a stock or a sector
stock-day	one stock on one date, the group we count news over

Text-Data Glossary (2) — Scoring, Methods, and Tasks

These are the words for turning text into a number, and the wider family of text tasks.

Term	Plain meaning
sentiment	the tone of a text — positive, negative, or neutral
lexicon	a fixed list of words — a *sentiment* lexicon also tags each word's tone
sentiment score	a number rating that tone (higher means more positive)
sentiment analysis	scoring the tone of text, the main task of this course
rules-based method	extract information with written rules (keywords, patterns)
dictionary-based method	score tone by matching words to a sentiment lexicon — **VADER** and **TextBlob** are two tools
lemmatisation	reduce a word to its dictionary form (`running` $\rightarrow$ `run`)
named entity recognition	find the names of people, firms, and places in text
topic modelling	group documents by the themes they discuss

Structured vs. Unstructured Data



Industry estimates (Gartner, IDC) put unstructured data at roughly **80%** of all business data.

Example — Statements and Filings

A Federal Reserve statement and a company 10-K are both unstructured text, and both move markets the moment they are released.

FEDERAL RESERVE press release

For release at 2:00 p.m. EDT June 12, 2024

Recent indicators suggest that economic activity has continued to expand at a solid pace. Job gains have remained strong, and the unemployment rate has remained low. Inflation has eased over the past year but remains elevated. In recent months, there has been modest further progress toward the Committee's 2 percent inflation objective.

The Committee seeks to achieve maximum employment and inflation at the rate of 2 percent over the longer run. The Committee judges that the risks to achieving its employment and inflation goals have moved toward better balance over the past year. The economic outlook is uncertain, and the Committee remains highly attentive to inflation risks.

In support of its goals, the Committee decided to maintain the target range for the federal funds rate at 5-1/4 to 5-1/2 percent. In considering any adjustments to the target range for the

FOMC policy statement (the Federal Reserve)

UNITED STATES
SECURITIES AND EXCHANGE COMMISSION
Washington, D.C. 20549

FORM 10-K

☒ ANNUAL REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

For the fiscal year ended January 28, 2024

OR

☐ TRANSITION REPORT PURSUANT TO SECTION 13 OR 15(d) OF THE SECURITIES EXCHANGE ACT OF 1934

Commission file number: 0-23985

NVIDIA

NVIDIA CORPORATION
(Exact name of registrant as specified in its charter)

Delaware	94-3177549
(State or other jurisdiction of incorporation or organization)	(I.R.S. Employer Identification No.)

NVIDIA 10-K annual report (a company filing)


THE TI

Max 18C, min 5C

Tuesday September 16 2008 timesonline.co.uk No 69430

Lehman collapse sends shockwave round world

hares and oil prices plunge, thousands lose jobs

are now uestionably in vorst financial since the t Depression'

The Times, 16 September 2008


Free delivery for six weeks

Get the Guardian sent to your home, with no delivery charge

Salmond cleared of sexual assault

Tuesday 24 March 2020

The Guardian

PM: 'Stay at home, this is a national emergency'

- Johnson drastically restricts movement to combat coronavirus
- Gatherings of more than two people in public to be prohibited
- Police given powers to disperse groups and enforce lockdown

Inside

The Guardian, 24 March 2020

Retail investors discuss stocks in public, in real time, on social platforms.

- X (formerly Twitter)
- Reddit, for example `r/wallstreetbets`
- StockTwits, and many others

Most platforms offer an application programming interface (API) that returns posts in real time, so you can score sentiment as the conversation happens. Reddit documents its own at `developers.reddit.com`.

How FinTech Uses Financial Text

FinTech use	Text source	What is calculated
News and social sentiment signals	articles and posts	a tone score for a trading or risk model
Earnings-call and filing analysis	transcripts and annual filings	topics, tone, and risk mentions
Robo-advice and support chat	customer messages	intent and an automated reply
Onboarding and risk screening	documents and news	name matches and risk flags

News tone has helped forecast market activity, so text can contain information relevant to prices (Tetlock, 2007). A core part of the project is using sentiment scores from text to build better portfolios.

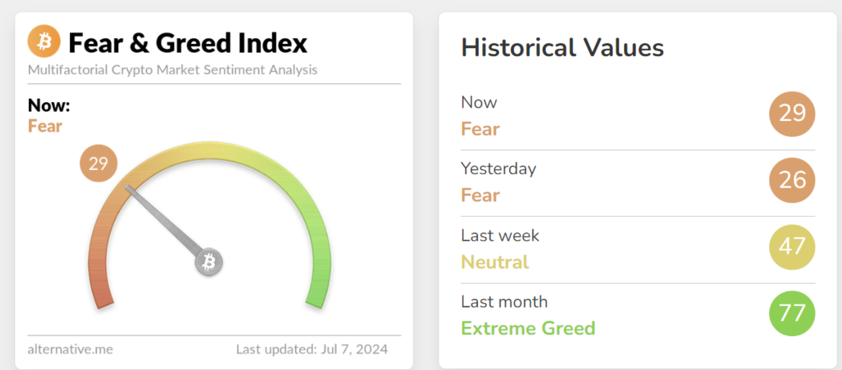
Two Ways to Score Text — Rules and Dictionaries

Machine learning with text data is challenging and beyond the scope of this course. There are easier ways to derive sentiment scores without training a model.

	Rules-based	Dictionary-based
How	written rules and keyword patterns	match words to a scored word list
Example	flag any sentence containing "profit" or "loss"	our mini lexicon, Loughran–McDonald
Training data	none	none
Strength	easy to implement	fast, scales to millions of documents
Weakness	lacks context awareness	lacks context awareness

The two are not rivals, and the Week 8 tools are both at once. **VADER** is, in its authors' words, "a lexicon and rule-based sentiment analysis tool" — a rated word list *plus* rules for negation, intensifiers, punctuation, and capitalisation (Hutto and Gilbert, 2014). **TextBlob**'s default scorer is built the same way, on the Pattern word list (De Smedt and Daelemans, 2012). Neither is trained. Week 7's count is dictionary-only, so it reads "not strong" as positive. The rules are what fix that.

A Real Sentiment Index — Crypto Fear and Greed



The **Fear and Greed Index** turns market sentiment into one number from 0 (extreme fear) to 100 (extreme greed), updated daily. Building an index like this from text is the end goal, and the focus of Week 8.

Why Use Text? From Words to a Trading Signal

raw headlines (text) → sentiment score per headline → daily stock and sector sentiment index → tilt portfolio weights

- **Part A (today).** Clean and describe the headlines into a tidy panel — one row per stock-day, text preserved, counts attached.
- **Part B (Week 8, Station 3).** Score that text with **VADER** and **TextBlob**, average into daily sentiment indices, and tilt positions with them.

Part A hands Station 3 a clean, ready panel, so that Part B can turn it into a stock-date sentiment score that tilts the portfolio.

Your Project News Headlines

- The project gives you one file of news **headlines** for 50 stocks across 10 sectors, dated 2020 to 2023.
- **Part A** of the project is Stages 1 and 2 of the Data Factory Floor, which is today's work — you load the headlines and describe them.
- **Part B** adds Stage 3, where you score sentiment with VADER and TextBlob, beginning in Week 8.

Today you load the project headlines and describe them, conducting basic feature engineering. You do not score sentiment yet.

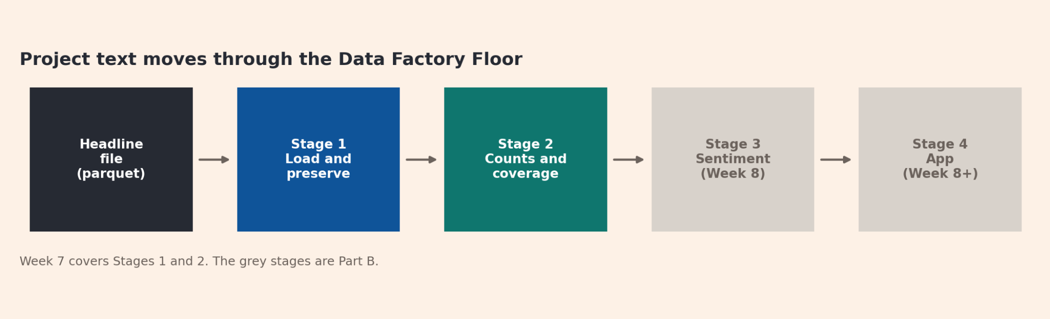
From the Practice Data to the Project Data

The practice exercise used a market-wide news file with one row per day. The project data is richer because every headline is tagged with the stock it is about.

	Practice data	Project data
One row	one trading day of news	one headline for one stock
Text column	`news_text`	`title`
Grouping	market direction	stock and sector
Rows	2,111	149,683

The steps are the same. The stock tag is what lets us describe the news one company at a time.

The Project Runs the Same Data Factory Floor



- Stage 1 loads the headline file and preserves the text.
- Stage 2 counts words and measures coverage.
- Stage 3 and Stage 4 are Part B. Week 8 scores sentiment on the same text.

Set Up the Run-Along Folder

Download the Week 7 folder (`week07_student_folder.zip`) from Ed and unzip it into your `fins2026/` folder. The scripts *and* the data are already inside, at

`fins2026/week7/project_text_dff/`

1. Open the folder in PyCharm and run scripts 01 to 04 in order.
2. Each script writes tables and figures to `output/`.

Nothing to copy and no path to set. `news_headlines.parquet` sits beside the scripts, and they read it from there. Scripts 01 to 03 build the tables. Script 04 draws the figures.

Load the Headlines Your Project Provides

In your project you load the file through the supplied helper.

```
from data_access import load_news_headlines
headlines = load_news_headlines()
```

For this run-along the scripts read the very same `news_headlines.parquet`, shipped beside them in the folder, so the numbers here match your project numbers.

Package	Job today
`pandas`	read the file and group rows
`matplotlib`	draw the coverage and top-term figures

Stage 1
Load and Preserve

One Headline Is One Row

The **unit of observation** is one news headline about one stock on one date.

Column	Example value
`date`	2020-01-01
`ticker`	ABBV
`sector`	Healthcare
`title`	Is AbbVie's Stock a Buy?
`url`	nasdaq.com/articles/is-abbvies-stock-a-buy...
`publisher`	often empty

Script 01 Loads and Checks the File

```
headlines = load_project_headlines()
print(headlines.shape)                    # (149683, 6)
print(headlines["ticker"].nunique())      # 50
print(headlines["sector"].nunique())      # 10
```

Check	Value
Headline rows	**149,683**
Stocks	50
Sectors	10
Date range	2020-01-01 to 2023-12-31

Check the Data Before You Trust It

Two checks run before any counting.

- **Missing values.** The core columns date, ticker, sector, title, and url have none. The publisher column is empty in **140,255** of 149,683 rows.
- **Exact duplicates.** Only 23 rows repeat every field. These differ from a headline that is reused for several stocks, which we keep on purpose.

Preserve the Headline Text

The project text column is `title`. We copy it to `text_raw` and never edit `text_raw`.

```
validated["text_raw"] = validated["title"]
assert (validated["text_raw"] == validated["title"]).all()
```

Week 8 scores sentiment on this exact text, so the capitalisation and punctuation must survive. Any word-count preparation goes in a separate column.

Keep one preserved copy of the headline, and build every feature in new columns beside it.

Stage 2
Describe the Text

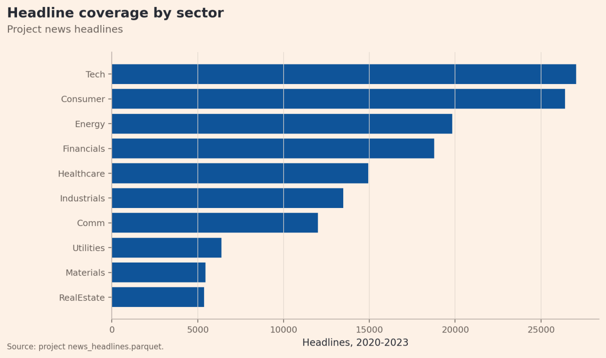
Stage 2 Describes the Text Without Scoring It

Part A stops before sentiment. In Stage 2 we answer three descriptive questions.

1. How much news does each stock and each sector have?
2. How long is a headline, and how many headlines fall on one day?
3. Which words appear most often for a stock or a sector?

None of these is a sentiment score. They describe the coverage so that a later score can be read in context.

News Coverage Is Uneven Across Sectors

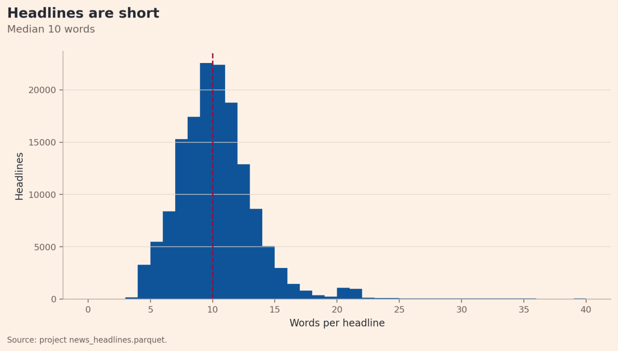


Tech and Consumer have the most headlines, near 27,000 each. Real estate and Materials have about 5,400. A sector with less news gives any later index a thinner daily signal.

News Coverage Is Uneven Across Stocks

NVDA has the most headlines at 9,025. The least-covered stock, AEP, has 661. A busy stock is not a more important one, so read raw counts beside coverage.

Headlines Are Short



A headline holds a median of **10** words. We store this as `n_words`, a count feature that sits beside `text_raw` and does not change it.

Most Days a Stock Has Only a Few Headlines

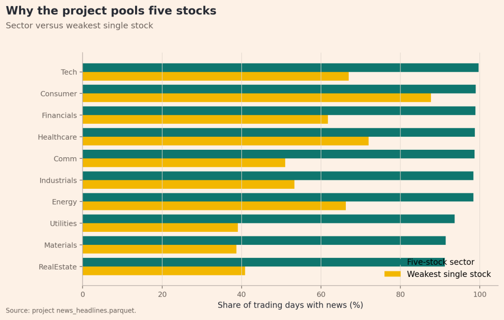
Group the rows by stock and date to get **stock-day** counts.

Measure	Value
Stock-day rows	**44,859**
Mean headlines per stock-day	3.34
Median headlines per stock-day	2

Two headlines on a typical day is thin, and on many days a single stock has no news at all.

One Stock Has News on Only 72% of Trading Days

On the other days that stock has no headline at all, so any measure built from its news alone has gaps.



One stock averages news on **72%** of trading days, and the worst-covered stock manages only 39%. Pooling the five stocks in a sector lifts that to **91 to 99.7%**. This is why the project builds each sector index from five stocks rather than one.

Script 03 splits each headline into word tokens and counts them.

```
def tokenize(text):
    return re.findall(r"[a-z][a-z']+", text.lower())

counts = Counter(word for title in titles for word in tokenize(title)
                 if word not in STOP_WORDS)
```

We remove a short list of **stop words** such as the, is, and of, but only in this count step. The preserved `text_raw` keeps every word.

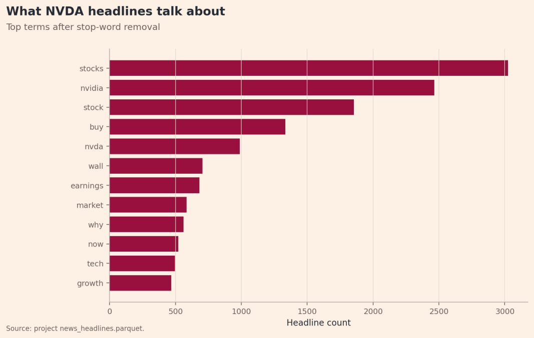
Frequent Words

Read down the list. Not one of these words names a company, and not one says what happened. Every one of them could sit in a headline about any of the 50 stocks.

Word	Count	Word	Count
stocks	32,058	dividend	7,631
stock	21,626	etf	7,354
buy	16,031	why	7,052
earnings	12,284	top	5,892
market	8,011	energy	5,717

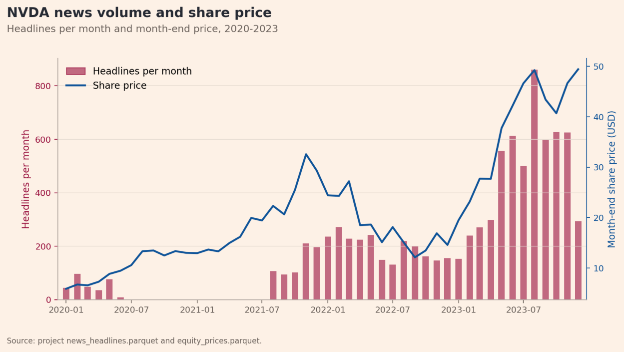
They are frequent because of how these headlines are written. **6,719** of them contain the phrase "Stocks to Buy". **10.9%** open with a digit, as in *7 Healthcare Stocks to Buy or Sell As Pricing Pressures Mount*, and **9.5%** end in a question mark, as in *Why Neurocrine Biosciences Soared 50.5% in 2019*. So a word count first tells you what kind of text you are holding — here, retail stock commentary — and only then anything about one company.

Examining NVIDIA, NVDA



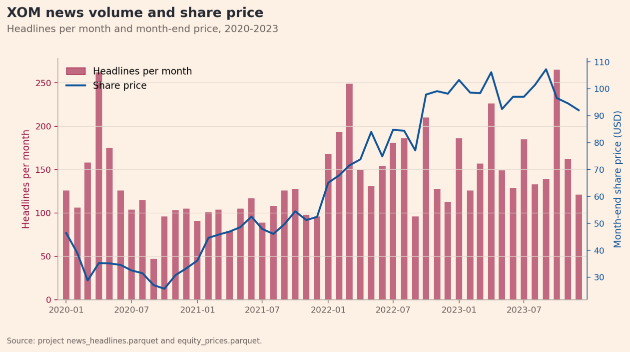
NVDA has 9,025 headlines and a vocabulary of 6,401 distinct words. The generic words stocks and buy still lead, but **nvidia** and **nvda** are the words that mark the coverage as being about this company.

News Volume and Share Price Rise Together



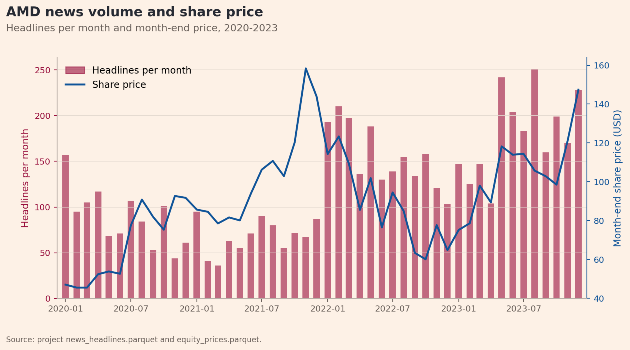
The bars show NVDA headlines per month. The line shows its month-end share price, which rose from about \$6 to \$49 across the sample, and both climb steeply through 2023. That co-movement is only a description, and it does not show that news moves the price, or that either one predicts future returns.

Exxon News and Price Also Rose Together



Exxon's price fell in the 2020 oil crash, then rose with the 2021 to 2022 energy boom from about $30 to over $100. News volume climbed alongside it. The same news-and-price pattern appears in a different sector.

AMD News Stayed High as the Price Fell in 2022



AMD's price and news both surged into late 2021. During the 2022 selloff the price fell by more than half while news volume stayed high. News volume follows attention, which rises on both good and bad news, so it does not tell you the direction of the price.

Different Sectors — Different Vocabulary

The same top-term feature shows Energy and Tech headlines are about different things.

Energy word	Count	Tech word	Count
oil	3,321	nvidia	3,535
energy	2,940	amd	2,981
exxon	1,570	intel	2,312
chevron	1,512	earnings	2,137

Oil and company names separate Energy from Tech. This describes the language, not which sector performs better.

Profiling All Stocks

Script 03 saves one summary row per stock, so the narrative scales to all 50.

Stock	Sector	Headlines	Mean words	Top word
NVDA	Tech	9,025	9.71	stocks
DIS	Consumer	8,617	9.11	disney
WMT	Consumer	7,230	9.00	walmart
INTC	Tech	6,286	9.41	intel
GS	Financials	5,644	9.68	goldman

For DIS, WMT, INTC, and GS the top word is the company name. For NVDA the generic word stocks still leads, so read past it to nvidia.

What These Counts Can and Cannot Say

Supported today	Not supported today
NVDA has the most headlines	NVDA is the best stock
Energy headlines mention oil	the news is positive or negative
coverage is thin on some days	word counts predict returns

A frequent word is not automatically important or positive. Tone needs a sentiment model, which is the Week 8 task. The word list you score with matters too — "liability" and "cost" sound negative in ordinary English but are everyday words in finance (Loughran and McDonald, 2011).

Stage 2 Produces the Text Foundation for Part B

Each script saves an output that your Part B work reads.

Saved output	What it holds
`01_validated_headlines`	the checked rows with preserved `text_raw`
`02_coverage_by_sector`	how many trading days each sector covers
`03_per_stock_summary`	counts, length, and top word for all 50 stocks

In Week 8, Part B scores `text_raw` with VADER and TextBlob, then averages the scores into daily stock and sector indices.

What You Can Now Do

- Load the project headlines and confirm the 50 stocks, 10 sectors, and 2020 to 2023 window.
- Check missing values and duplicates, and preserve the `title` text.
- Count headlines by stock, by sector, and by day, and measure coverage.
- Build word counts and top-term profiles for one stock and for a sector.
- State what a count describes and what it cannot claim.

Part A is done for the text data, and every row now carries a checked source, preserved text, and descriptive features ready for Week 8.

References I

Tom De Smedt and Walter Daelemans. Pattern for Python. *Journal of Machine Learning Research*, 13:2063–2067, 2012.

Clayton J. Hutto and Eric Gilbert. VADER: A parsimonious rule-based model for sentiment analysis of social media text. *Proceedings of the International AAAI Conference on Web and Social Media*, 8(1):216–225, 2014. doi: 10.1609/icwsm.v8i1.14550.

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